

True or false?

Computers decide what to do by asking questions and determining whether the answers are true or false. Questions that only have two possible answers are called “Boolean expressions”.

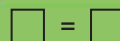
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Comparing numbers

You can compare numbers using the “=” block from the “Operators” section of the blocks palette.



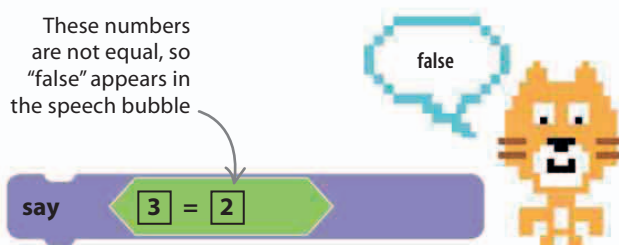
◁ The “=” block

This block will give one of two answers: “true” if the two numbers in the boxes are equal, and “false” if they aren’t.

The numbers are equal, so “true” appears in the speech bubble



These numbers are not equal, so “false” appears in the speech bubble



△ True answer

Using an “=” block inside a speech block will make “true” or “false” appear in a sprite’s speech bubble.

△ False answer

If the numbers in the block are different, the sprite’s speech bubble will contain the word “false”.

Comparing variables

You can use variables inside comparison blocks. It’s not worth comparing fixed numbers because the result will always be the same, whereas the value of variables can change.

set **age** ▼ to **10**

△ Create a variable

Click the “Data” button in the blocks palette and create a new variable called “age”. Set its value to 10 (click on the block to make sure the value has changed). Drag the “age” variable into the comparison blocks.

age

= 7

This sign means “equals”, so the block is asking if “age” is equal to 7. The answer here is “false”, because “age” is 10

age

> 11

This sign means “more than”, so the block is asking if “age” is greater than 11. The answer is “false”, because 10 is not bigger than 11

age

< 18

This sign means “less than”, so the block is asking if “age” is less than 18. The answer will be “true”, as 10 is smaller than 18

△ Comparing numbers

Find the green comparison blocks in the “Operators” menu. As well as checking whether two numbers are equal, you can check whether one is greater or less than another.